

ShieldGig — Phase 1 Research Document

Deadline: Fill everything before Day 3 Rule: Every fact needs a source link. No guessing.

HOW TO USE THIS DOCUMENT

- Each section has WHAT TO SEARCH and WHAT TO FIND
 - Copy the search queries directly into Google
 - Paste your findings under each section
 - Add the source link next to every fact
 - If you can't find something after 15 minutes, write "NOT FOUND" and move on
-

SECTION 1 — Parametric Insurance Basics

Assigned to: P1 Time: 2 hours

What To Search

```
"parametric insurance explained simply"
"parametric insurance vs traditional insurance"
"parametric insurance India examples"
"parametric crop insurance India PMFBY"
"parametric insurance payout trigger examples"
```

What To Find

1. Simple definition of parametric insurance (in your own words after reading):

█ [FILL THIS IN]

2. How is it different from traditional insurance? (3 key differences):

█ Difference 1: [FILL THIS IN]
█ Difference 2: [FILL THIS IN]
█ Difference 3: [FILL THIS IN]

3. Real examples of parametric insurance that already exist in India:

█ Example 1: [FILL THIS IN] — Source: [LINK]
█ Example 2: [FILL THIS IN] — Source: [LINK]
█ Example 3: [FILL THIS IN] — Source: [LINK]

4. What is a "trigger threshold" in parametric insurance?

| [FILL THIS IN] — Source: [LINK]

5. What is a "payout index" and how is it calculated?

| [FILL THIS IN] — Source: [LINK]

6. Why can't parametric insurance be used for health/life? (Important for judges)

| [FILL THIS IN] — Source: [LINK]

7. What is IRDAI's stance on parametric/microinsurance products?

| Search: "IRDAI microinsurance guidelines 2024"
| [FILL THIS IN] — Source: [LINK]

Key Insight For README:

| After research, write 2-3 sentences explaining why parametric insurance is the RIGHT model for gig workers specifically:
| [FILL THIS IN]

SECTION 2 — Gig Worker Income Data India

Assigned to: P2 Time: 2 hours

What To Search

"Zomato delivery partner income India 2024"
"gig worker monthly income India statistics"
"how much does swiggy delivery partner earn per week"
"gig economy workers India 2024 report"
"platform gig workers India income loss disruption"
"NITI Aayog gig workers India report"

What To Find

1. How many platform gig delivery workers are there in India?

| Number: [FILL THIS IN] — Source: [LINK]

2. Average weekly income of a Zomato/Swiggy delivery partner:

| Amount: ₹[FILL THIS IN]/week — Source: [LINK]

3. Average number of deliveries per day:

| Number: [FILL THIS IN] deliveries/day — Source: [LINK]

4. Average earning per delivery:

| Amount: ₹[FILL THIS IN]/delivery — Source: [LINK]

5. What percentage of income do gig workers lose due to disruptions?

Percentage: [FILL THIS IN]% — Source: [LINK]

(Note: Use case document says 20-30% — find a source that confirms this)

6. Do gig workers have any existing financial safety net?

[FILL THIS IN] — Source: [LINK]

7. What are the peak earning hours for delivery workers?

Morning peak: [FILL THIS IN]

Evening peak: [FILL THIS IN]

Source: [LINK]

8. How many working days per week does an average delivery partner work?

Days: [FILL THIS IN] — Source: [LINK]

9. What type of phones do delivery workers use? (Important for Flutter performance argument)

Search: "average smartphone price gig workers India"

[FILL THIS IN] — Source: [LINK]

10. NITI Aayog report key findings on gig workers:

Search: "NITI Aayog India's Booming Gig and Platform Economy report"

Finding 1: [FILL THIS IN]

Finding 2: [FILL THIS IN]

Source: [LINK]

Key Insight For README:

Write the most powerful statistic you found that proves gig workers need income protection:

[FILL THIS IN]

SECTION 3 — Weather & Disruption Data Chennai

Assigned to: P3 Time: 1.5 hours

What To Search

"Chennai average annual rainfall mm"

"Chennai northeast monsoon months rainfall data"

"Chennai rain days per year"

"IMD Chennai rainfall classification heavy very heavy"

"Chennai flood history years"

"Chennai cyclone history Bay of Bengal"

"Cyclone Michaung Chennai impact 2023"

"Chennai AQI average pollution levels"

"Chennai traffic congestion index"

What To Find

1. How many rainy days does Chennai get per year on average?

| Days: [FILL THIS IN] — Source: [LINK]

2. Northeast monsoon months in Chennai:

| Months: [FILL THIS IN] — Source: [LINK]

3. Average rainfall during peak monsoon month (mm):

| Amount: [FILL THIS IN]mm — Source: [LINK]

4. IMD rain classification thresholds (CRITICAL for our parametric triggers):

| Light rain: [FILL THIS IN]mm
| Moderate rain: [FILL THIS IN]mm
| Heavy rain: [FILL THIS IN]mm
| Very heavy rain: [FILL THIS IN]mm
| Extremely heavy rain: [FILL THIS IN]mm
| Source: [LINK]

5. How many days per year does Chennai exceed "Heavy Rain" threshold?

| Days: [FILL THIS IN] — Source: [LINK]

6. Cyclone Michaung 2023 — how many days did it disrupt Chennai?

| Days: [FILL THIS IN] — Source: [LINK]
| Estimated income loss for workers: [FILL THIS IN] — Source: [LINK]

7. Chennai average AQI levels:

| Annual average AQI: [FILL THIS IN]
| Worst month AQI: [FILL THIS IN]
| Source: [LINK]

8. Does Chennai ever exceed AQI 200 (threshold for our trigger)?

| Yes/No: [FILL THIS IN] — Source: [LINK]

9. Chennai waterlogging zones (which areas flood first?):

| Zone 1: [FILL THIS IN]
| Zone 2: [FILL THIS IN]
| Zone 3: [FILL THIS IN]
| Source: [LINK]

10. OpenWeatherMap API — free tier limits:

Search: "OpenWeatherMap free tier API calls per day"

Calls per day free: [FILL THIS IN] — Source: [LINK]

Key Insight For README:

Write one sentence explaining why Chennai is the perfect pilot city for ShieldGig:

[FILL THIS IN]

SECTION 4 — Fraud Detection Methods

Assigned to: P4 Time: 1.5 hours

What To Search

```
"insurance fraud detection methods AI"
"parametric insurance fraud risks"
"location spoofing insurance fraud"
"anomaly detection insurance claims"
"gig worker fraud detection GPS"
"isolation forest fraud detection explained"
"behavioral analytics insurance fraud"
```

What To Find

1. What are the most common types of insurance fraud in India?

Type 1: [FILL THIS IN] — Source: [LINK]

Type 2: [FILL THIS IN] — Source: [LINK]

Type 3: [FILL THIS IN] — Source: [LINK]

2. What percentage of insurance claims are fraudulent in India?

Percentage: [FILL THIS IN]% — Source: [LINK]

3. How much does insurance fraud cost the industry annually in India?

Amount: ₹[FILL THIS IN] — Source: [LINK]

4. What is location spoofing and how do fraudsters use it?

[FILL THIS IN] — Source: [LINK]

5. What is anomaly detection in fraud context? (simple explanation)

[FILL THIS IN] — Source: [LINK]

6. What is Isolation Forest algorithm? (simple explanation)

Search: "isolation forest algorithm fraud detection simple explanation"

[FILL THIS IN] — Source: [LINK]

7. How does behavioral baseline fraud detection work?

| [FILL THIS IN] — Source: [LINK]

8. What fraud signals does Zomato/Swiggy use to detect fake delivery partners?

| Search: "Zomato fake delivery fraud detection"
| [FILL THIS IN] — Source: [LINK]

9. What is "claim velocity" in insurance fraud?

| [FILL THIS IN] — Source: [LINK]

10. What techniques do real insurers use to detect parametric insurance fraud?

| Search: "parametric insurance fraud prevention techniques"
| [FILL THIS IN] — Source: [LINK]

Key Insight For README:

| Write one sentence explaining why fraud detection is especially important in parametric insurance vs traditional insurance:
| [FILL THIS IN]

SECTION 5 — Guidewire Products & Business

Assigned to: P5 Time: 1.5 hours

What To Search

"Guidewire Software what does it do"
"Guidewire PolicyCenter explained"
"Guidewire ClaimCenter explained"
"Guidewire BillingCenter explained"
"Guidewire customers India insurance companies"
"Guidewire insurance platform gig economy"
"Guidewire annual revenue 2024"
"Guidewire cloud platform insurtech"
"Guidewire DEVTrails previous winners"

What To Find

1. What does Guidewire do in one sentence?

| [FILL THIS IN] — Source: [LINK]

2. What are Guidewire's three main products?

| Product 1: [FILL THIS IN] — what it does: [FILL THIS IN]
| Product 2: [FILL THIS IN] — what it does: [FILL THIS IN]

Product 3: [FILL THIS IN] — what it does: [FILL THIS IN]

Source: [LINK]

3. Which Indian insurance companies use Guidewire?

Company 1: [FILL THIS IN]

Company 2: [FILL THIS IN]

Source: [LINK]

4. What is Guidewire's revenue / how big is the company?

Revenue: \$[FILL THIS IN] — Source: [LINK]

5. What is Guidewire's main value proposition to insurance companies?

[FILL THIS IN] — Source: [LINK]

6. What is Guidewire Marketplace?

[FILL THIS IN] — Source: [LINK]

7. Has Guidewire done anything related to gig economy or microinsurance?

[FILL THIS IN] — Source: [LINK]

8. What does Guidewire look for in innovation / hackathon projects?

Search: "Guidewire innovation hackathon insurtech"

[FILL THIS IN] — Source: [LINK]

9. How does ShieldGig align with Guidewire's ClaimCenter product specifically?

(Think about this after reading about ClaimCenter — write your own analysis)

[FILL THIS IN]

10. How does ShieldGig align with Guidewire's PolicyCenter product?

(Think about this after reading about PolicyCenter — write your own analysis)

[FILL THIS IN]

Key Insight For Pitch:

Write the single most important thing you learned about Guidewire that should change how we present ShieldGig:

[FILL THIS IN]

SECTION 6 — Competitor Insurance Products

Assigned to: P1 (Day 2) Time: 1.5 hours

What To Search

"gig worker insurance India 2024"
"delivery partner insurance Zomato Swiggy"
"platform worker income protection insurance India"
"parametric insurance gig workers global"
"microinsurance gig economy startup India"
"Ola Uber driver insurance India"
"ICICI Lombard gig worker insurance"
"Digit insurance gig workers"
"SafeWings insurance delivery partners"

What To Find

1. Does Zomato offer any insurance to its delivery partners currently?

Yes/No: [FILL THIS IN]
What it covers: [FILL THIS IN]
What it does NOT cover: [FILL THIS IN]
Source: [LINK]

2. Does Swiggy offer any insurance to its delivery partners?

Yes/No: [FILL THIS IN]
What it covers: [FILL THIS IN]
What it does NOT cover: [FILL THIS IN]
Source: [LINK]

3. Any startup in India building income protection for gig workers?

Company 1: [FILL THIS IN] — what they do: [FILL THIS IN] — Source: [LINK]
Company 2: [FILL THIS IN] — what they do: [FILL THIS IN] — Source: [LINK]

4. Any global startup doing parametric insurance for gig workers?

Company 1: [FILL THIS IN] — Source: [LINK]
Company 2: [FILL THIS IN] — Source: [LINK]

5. What gap exists in ALL current products that ShieldGig fills?

(Write this yourself after researching — this is the most important competitive insight)
Gap 1: [FILL THIS IN]
Gap 2: [FILL THIS IN]
Gap 3: [FILL THIS IN]

6. What is the total addressable market for gig worker insurance in India?

Search: "gig economy insurance market size India"
Size: ₹[FILL THIS IN] crore — Source: [LINK]

Key Insight For Pitch:

Write one sentence that explains why ShieldGig beats every existing product:
[FILL THIS IN]

SECTION 7 — Technical Research

Assigned to: P3 + P4 together Time: 1 hour

What To Search

```
"OpenWeatherMap API free tier documentation"  
"AQICN API key registration free"  
"Supabase free tier limits 2024"  
"Flutter background location Android tutorial"  
"flutter_background_service package documentation"  
"geolocator flutter package"  
"Render free tier Node.js deployment"  
"Firebase Cloud Messaging Flutter setup"
```

What To Find

1. OpenWeatherMap — how to get free API key:

Steps: [FILL THIS IN]
Free calls per day: [FILL THIS IN]
Does it give rainfall in mm/hr? Yes/No: [FILL THIS IN]
Source: [LINK]

2. AQICN API — how to get free key:

Steps: [FILL THIS IN]
Source: [LINK]

3. Flutter background location — which package to use:

Package name: [FILL THIS IN]
Does it work when screen is off on Android? [FILL THIS IN]
Source: [LINK]

4. Supabase — confirm free tier is enough:

DB size limit: [FILL THIS IN]
API calls limit: [FILL THIS IN]
Source: [LINK]

5. Render free tier — confirm it hosts Node.js:

Yes/No: [FILL THIS IN]
Any sleep/cold start issues? [FILL THIS IN]

SECTION 8 — Key Statistics For Video & Pitch

Assigned to: P2 (compile from all sections) Fill this LAST after all other sections are done

These are the numbers that go into the 2-minute video and final pitch deck. Every number must have a source.

Statistic	Number	Source
Gig delivery workers in India		
Average weekly income	₹	
Income lost to disruptions	%	
Rain days in Chennai per year	days	
Insurance fraud rate India	%	
Cost of fraud to industry India	₹ crore	
Gig insurance market size India	₹ crore	
Claims processing cost traditional	₹ per claim	
Cyclone Michaung disruption days	days	

RESEARCH COMPLETION CHECKLIST

Before Day 3 starts, confirm:

- ☐ Section 1 complete — P1
- ☐ Section 2 complete — P2
- ☐ Section 3 complete — P3
- ☐ Section 4 complete — P4
- ☐ Section 5 complete — P5
- ☐ Section 6 complete — P1 Day 2
- ☐ Section 7 complete — P3 + P4
- ☐ Section 8 compiled — P2
- ☐ All facts have source links
- ☐ Key insights written for each section
- ☐ Statistics table filled completely

RESEARCH RULES

1. **15-minute rule:** If you cannot find something in 15 minutes, write NOT FOUND and move on. Don't waste time.
2. **Source everything:** No fact without a link. Judges may ask where numbers came from.
3. **Use reliable sources:** Prefer government sites (IMD, IRDAI, NITI Aayog), news sites (Economic Times, Mint, The Hindu), and company official pages.
4. **No Wikipedia as final source:** Use it to understand concepts but find the original source it cites.
5. **Share in real time:** Use a shared Google Doc so everyone sees progress live.
6. **Flag conflicts:** If two sources give different numbers, write both and flag it — team decides which to use.